Bhoomika R Hegde

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PUBLICATIONS

IJDSA 2026.Transaction Graph-Based Predictive Hurdle Model for Credit Scoring in DeFi Lending Protocols.. **2026**

- Developed a two-stage credit-risk model combining wallet behaviour, transaction-graph and community-risk features across 140K+ transactions and 13K+ wallets. The integrated score achieved AUC-ROC 0.8738; all 100 highest-risk wallets were liquidated in the evaluation period.

Canine EEG Validation before decoding..

- Developed a reproducible EEG validation framework spanning data integrity, QC, confound analysis, and subject-independent decoding across 16 dogs and 2,820 epochs, with calibrated null testing to prevent overstated ML claims.

ICMLC 2026.Discrete Representation Learning: A Review.. **2026**

- Reviewed discrete representation learning methods across VQ-VAE, product quantization, and discrete latent models, comparing architectures, training objectives, and applications in generative modeling.

PROJECTS

Dog Activity Recognition System

- Architected an event-driven ML platform for multimodal activity recognition using Python, Kafka, FastAPI, PostgreSQL, and Docker.
- Engineered robust inter-service communication through API contracts, Kafka schemas, retries, and end-to-end traceability.
- Integrated CLIP-based zero-shot inference with REST APIs and production-ready testing to enable reliable model deployment.

Image Representation Enhancement using ViT and Product Quantisation

- Identified limitations in VQ-VAE2 and proposed a PQ-VAE architecture to improve representation learning and codebook efficiency.
- Designed an end-to-end encoding pipeline combining Vision Transformers with structured latent decomposition and differentiable codebook learning.
- Improved reconstruction quality and compression efficiency over baseline models, validated using SNR and SSIM; contributed to a companion paper accepted at ICMLC 2026.

Canine EEG Validation Framework

- Built a validation-first canine EEG pipeline for 17 recordings and 2,820 epochs, validating signal import, events, labels, QC, and confounds.
- Implemented leave-one-dog-out decoding using 88 EEG features and logistic regression, achieving macro AUC = 0.5034 against a within-dog permutation null.
- Developed reproducible validation reports with checksums, configuration hashes, seeds, and software-version tracking.

EXPERIENCE

WLDD | ***ML Engineer - I*** **Aug 2026 - Present**

- Built a creator–campaign matching system using creator content and profile signals. On a 135-brief benchmark, improved Top-1 accuracy from 83% to 90% and Top-5 from 80% to 91%, reducing missed matches from 9 to 1.
- Compared dense, keyword and hybrid retrieval, then tuned ranking and LLM reranking. A confidence check skips reranking on 34% of queries without reducing accuracy.
- Built creator-data enrichment for niche, audience, brand, location and language; added deduplication, failure tracking and resumable backfills. Added contact extraction and search filters, keeping lookup private until authentication was available.

Yugasys | ***AI Engineer Intern*** **Feb 2026 - Jun 2026**

- Built a voice-AI backend linking speech-to-text, intent parsing and application actions across 10+ flows, enabling voice-driven task completion.
- Implemented WebSocket/Agora communication with connection-state management, retries and heartbeat checks to handle unstable networks.
- Integrated agent services through MCP with defined service boundaries and event contracts; resolved cross-service failure cases while keeping CI/CD stable.

Zeru | ***AIML Engineer Intern*** **Jun 2025 - Aug 2025**

- Built a modular data processing pipeline (Python) spanning 5 blockchain protocols for bot detection, enabling protocol-specific preprocessing components to evolve independently
- Implemented end-to-end traceability across the pipeline, tagging each transaction from ingestion through classification to ensure debuggability and auditability
- Improved production readiness by reducing silent failure modes, adding validation checks at stage boundaries, and ensuring error states were logged and observable

PES University | ***Teaching Assistant-Advanced Data Analytics (Dr. Bhaskar Jyothi Das)*** **Aug 2025 - Dec 2025**

- Write exercises and case studies for a 250-student course; also sit with student teams when they're stuck on architecture decisions or debugging something that doesn't make sense yet.
- Mentored student teams on project architecture, debugging strategies, and efficient algorithm implementation.
- Helped standardise how work gets graded so assessment doesn't drift across 250 submissions.

Grain For All | ***Engineer Intern*** **Aug 2024 - Nov 2024**

- Developed a Python-based ETL system to parse and normalize EPA documents into structured outputs using a modular plugin architecture for inconsistent formats
- Integrated a multi-step reasoning pipeline using ReAct and AutoGen; added validation and logging at each stage to prevent silent data corruption
- Collaborated in a production environment with cross-functional teams, ensuring modular updates without breaking downstream pipeline components

EDUCATION

PES University **2022 - 2026**

B.Tech, Computer Science and Engineering (AI & ML)

- **Achievements:** DAC scholarships, Head of Sponsorship Aatmatrisha'25, Head of Sponsorship Chords'24, Head of Social Media AIKYA